B.TECH FIRST YEAR | ENGINEERING PHYSICS

UNIT II

Crystallography & X-ray Diffraction

Complete Study Module

Theory · Derivations · Diagrams · Solved Questions · Quick Revision

Prepared as per University Curriculum

References: Avadhanulu & Kshirsagar · D.K. Bhattacharya · B.K. Pandey

Covers: Space Lattice · Bravais Lattices · Miller Indices · Bragg's Law · X-ray Diffractometer · Laue & Powder Methods

Table of Contents

Section 1 | Introduction to Crystallography

1.1 Definition

Crystallography is the scientific study of the arrangement of atoms in crystalline solids. It deals with the geometry, symmetry, physical properties, and the methods used to determine the internal structure of crystals. The term originates from the Greek words *krystallos* (clear ice) and *graphia* (writing/description).

In engineering physics, crystallography provides a foundation for understanding the mechanical, electrical, optical, and thermal properties of materials. The precise arrangement of atoms governs how a material responds to external stimuli such as stress, heat, electric fields, and electromagnetic radiation.

1.2 Classification of Solids

Solids are broadly classified into two categories based on the regularity of atomic arrangement:

Property	Crystalline Solids	Amorphous (Non-crystalline) Solids
Atomic Arrangement	Long-range periodic order	Short-range order only
Melting Point	Sharp and well-defined	No sharp melting point
Anisotropy	Anisotropic (direction-dependent)	Isotropic
Examples	NaCl, Cu, Si, Diamond	Glass, Rubber, Wax, Plastics
Cleavage	Along definite planes	Irregular fracture

1.3 Importance in Engineering

- Determines mechanical strength, ductility, and hardness of metals.
- Explains electrical conductivity and semiconductor behaviour.
- Governs the diffraction of X-rays, electrons, and neutrons used in non-destructive testing.
- Foundation for designing alloys, ceramics, thin films, and nanomaterials.
- Enables drug design via protein crystallography in biotechnology.

KEY POINT

All crystalline materials exhibit a *periodic three-dimensional arrangement* of atoms. This periodicity gives rise to the concept of a crystal lattice, which is the mathematical framework underlying crystallography.

1.4 Historical Background

Nicolas Steno (1669) observed constant interfacial angles in quartz crystals. René-Just Haüy (1784) proposed the structural unit cell concept. William Henry Bragg and William Lawrence Bragg (1913) established X-ray diffraction as a tool for crystal structure determination, for which they were awarded the Nobel Prize in Physics in 1915.

Section 2 | Space Lattice and Basis

2.1 Space Lattice – Definition

A **space lattice** (also called a crystal lattice or Bravais lattice) is defined as an infinite, periodic array of geometric points in three-dimensional space such that every point has an identical environment. In other words, if the lattice is translated by a lattice vector, the arrangement of points looks exactly the same.

Mathematically, a space lattice is described by three fundamental translation vectors **a**, **b**, and **c** (called primitive vectors or basis vectors of the lattice). Any lattice point can be expressed as:

LATTICE VECTOR

$R = n_1a + n_2b + n_3c$

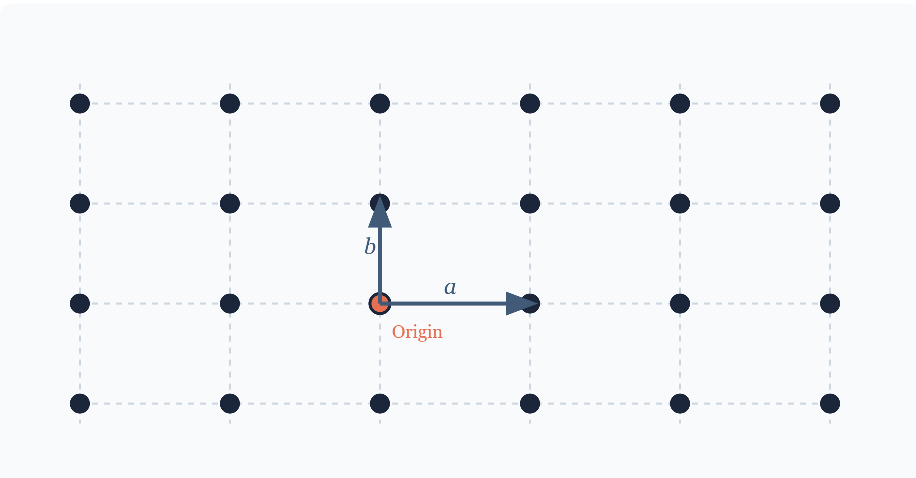
where n_1, n_2, n_3 are integers (positive, negative, or zero)

a, b, c are the primitive translation vectors

2.2 Properties of a Space Lattice

- Every lattice point is identical to every other lattice point.
- The arrangement of atoms about every lattice point is exactly the same.
- A space lattice is an abstract geometric concept; actual atoms are attached to each lattice point via the basis.
- There are exactly **14 distinct space lattices** in three dimensions (Bravais lattices).

Figure 2.1 – Two-dimensional space lattice showing periodic point arrangement



Each point has identical surroundings; *a* and *b* are primitive translation vectors.

2.3 Basis – Definition

A **basis** (also called a motif) is the atom or group of atoms associated with each lattice point. A basis can be a single atom (as in metals like copper) or a group of atoms (as in NaCl where each lattice point has one Na and one Cl atom).

The relationship between a crystal structure, space lattice, and basis is expressed as:

FUNDAMENTAL RELATION

```
Crystal Structure = Space Lattice + Basis

Every lattice point carries an identical basis
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2.4 Primitive Cell vs. Non-primitive Cell

Parameter	Primitive Cell	Non-primitive Cell
Lattice Points	Exactly 1	More than 1
Volume	Minimum possible	Larger (multiple of primitive)
Examples	Simple cubic primitive	BCC, FCC conventional cells
Also Called	Wigner-Seitz cell	Conventional unit cell

Section 3 | Unit Cell and Lattice Parameters

3.1 Unit Cell – Definition

A **unit cell** is the smallest repeating structural unit of a crystal lattice that, when translated repeatedly in all three directions, generates the entire crystal structure without gaps or overlaps. It is the fundamental building block of the crystal.

3.2 Lattice Parameters

A unit cell in three dimensions is completely described by six parameters, called **lattice parameters** (or crystallographic parameters):

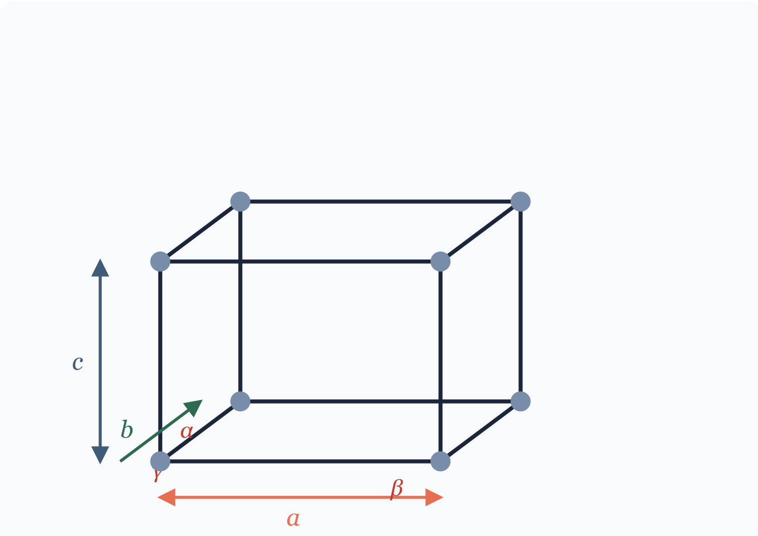
- **Three edge lengths:** a, b, c (lengths along the x, y, z crystallographic axes).
- **Three inter-axial angles:** α (angle between b and c axes), β (angle between a and c axes), γ (angle between a and b axes).

SIX LATTICE PARAMETERS

Edge lengths: a, b, c (in Angstroms, 1 Å = 10^{-10} m)

Inter-axial angles: α (b^c), β (a^c), γ (a^b)

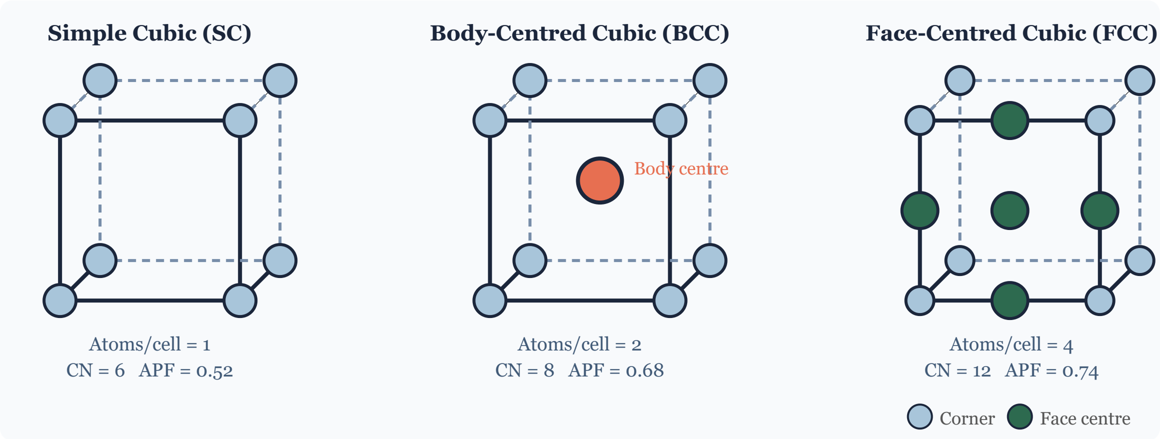
Figure 3.1 – Unit cell showing lattice parameters a, b, c and angles α, β, γ



Lattice parameters: edge lengths a, b, c and inter-axial angles α, β, γ

3.3 SC, BCC, and FCC Unit Cells

Figure 3.2 – Unit cells: (a) Simple Cubic | (b) Body-Centred Cubic | (c) Face-Centred Cubic



Solid circles represent atom positions; CN = coordination number; APF = atomic packing factor

Section 4 | Bravais Lattices

4.1 Definition and Concept

Auguste Bravais (1850) proved that there are exactly **14 distinct space lattices** (called Bravais lattices) that can fill three-dimensional space with identical geometric points having the same environment. These 14 lattices are grouped among the 7 crystal systems.

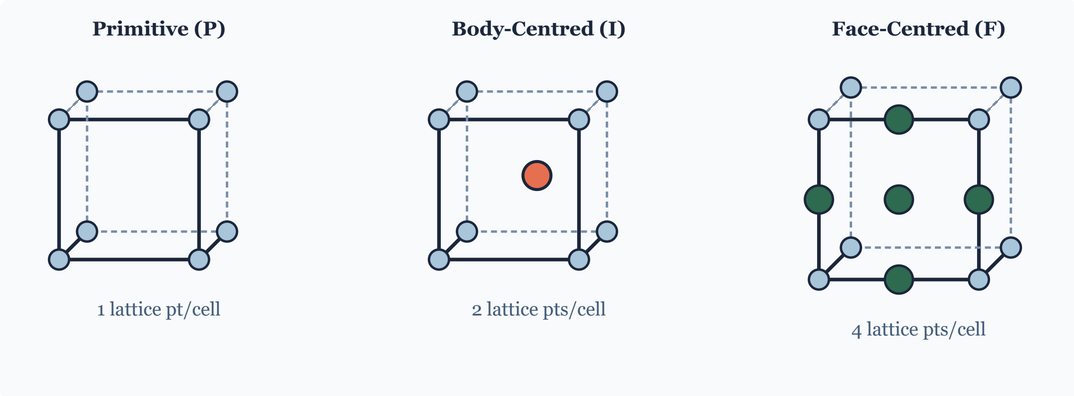
Each Bravais lattice is defined by the symmetry of the arrangement of lattice points. The 14 types arise from combining 7 primitive lattice types with four lattice centerings: Primitive (P), Body-centred (I), Face-centred (F), and Base-centred (C or A or B).

4.2 The 14 Bravais Lattices – Summary Table

Crystal System	Bravais Lattice Types	Number	Lattice Parameters
Cubic	P, I (BCC), F (FCC)	3	$a = b = c$; $\alpha = \beta = \gamma = 90°$
Tetragonal	P, I	2	$a = b \neq c$; $\alpha = \beta = \gamma = 90°$
Orthorhombic	P, I, F, C	4	$a \neq b \neq c$; $\alpha = \beta = \gamma = 90°$
Hexagonal	P	1	$a = b \neq c$; $\alpha = \beta = 90°$, $\gamma = 120°$
Rhombohedral (Trigonal)	P	1	$a = b = c$; $\alpha = \beta = \gamma \neq 90°$
Monoclinic	P, C	2	$a \neq b \neq c$; $\alpha = \gamma = 90°$, $\beta \neq 90°$
Triclinic	P	1	$a \neq b \neq c$; $\alpha \neq \beta \neq \gamma \neq 90°$
Total		**14**	

Note on Centering Symbols: P = Primitive (lattice points at corners only); I = Body-centred (+ 1 point at body centre); F = Face-centred (+ 1 point at each face centre); C = Base-centred (+ 1 point at each base face centre).

Figure 4.1 – Representative Bravais lattice types (cubic family shown)



The three cubic Bravais lattices: Primitive, Body-centred, and Face-centred

WHY ONLY 14 BRAVAIS LATTICES?

The restriction to 14 arises from the requirement that the lattice must fill space with translational periodicity AND maintain point-group symmetry. Not all combinations of 7 crystal systems and 4 centerings produce geometrically distinct lattices. For example, Face-centred tetragonal reduces to Body-centred tetragonal and is therefore not a new Bravais lattice.

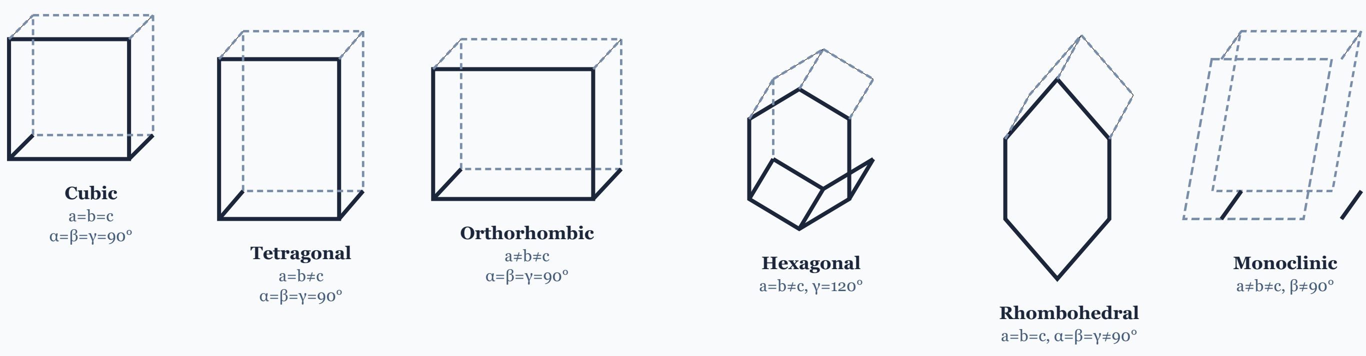
Section 5 | Crystal Systems in Three Dimensions

5.1 The Seven Crystal Systems

All crystalline substances belong to one of **seven crystal systems**, classified by the geometric relationships between the unit cell edge lengths (a, b, c) and the inter-axial angles (α, β, γ). The seven systems are listed below with their symmetry characteristics.

System	Axial Lengths	Axial Angles	Bravais Lattices	Examples
Cubic	a = b = c	α = β = γ = 90°	3 (P, I, F)	NaCl, Cu, Fe, Al
Tetragonal	a = b ≠ c	α = β = γ = 90°	2 (P, I)	SnO_2, TiO_2, In
Orthorhombic	a ≠ b ≠ c	α = β = γ = 90°	4 (P,I,F,C)	$BaSO_4$, S, I_2
Hexagonal	a = b ≠ c	α = β = 90°, γ = 120°	1 (P)	Mg, Zn, graphite
Rhombohedral	a = b = c	α = β = γ ≠ 90°	1 (P)	Calcite, Bi, Sb
Monoclinic	a ≠ b ≠ c	α = γ = 90°, β ≠ 90°	2 (P, C)	$CaSO_4{\cdot}2H_2O$, monoclinic S
Triclinic	a ≠ b ≠ c	α ≠ β ≠ γ ≠ 90°	1 (P)	$CuSO_4{\cdot}5H_2O$, $K_2Cr_2O_7$

Figure 5.1 – Schematic representation of the seven crystal systems



Summary: 7 Crystal Systems → 14 Bravais Lattices

The most symmetric: Cubic | Least symmetric: Triclinic

Cubic (3) + Tetragonal (2) + Orthorhombic (4) + Hexagonal (1) + Rhombohedral (1) + Monoclinic (2) + Triclinic (1) = 14

Schematic unit cell shapes for six of the seven crystal systems (Triclinic has no right angles or equal lengths)

Section 6 | Coordination Number and Atomic Packing Factor

6.1 Coordination Number – Definition

The **coordination number (CN)** of an atom in a crystal structure is defined as the number of nearest-neighbour atoms directly touching or in contact with that atom. It measures how tightly packed the atoms are around any given atom. A higher coordination number generally indicates a more closely packed structure.

6.2 Atomic Packing Factor (APF) – Definition

The **Atomic Packing Factor** (also called Packing Efficiency or Packing Fraction) is defined as the fraction of the total volume of the unit cell that is actually occupied by atoms. Atoms are assumed to be hard, non-compressible spheres.

APF GENERAL FORMULA

APF = (Number of atoms per unit cell × Volume of one atom) / Volume of unit cell

$APF = (Z \times (4/3)\pi r^3) / a^3$ [for cubic systems]

where Z = effective number of atoms/cell, r = atomic radius, a = lattice parameter

6.3 APF for Simple Cubic (SC) Structure

In a simple cubic structure, atoms are located only at the eight corners of the cube. Each corner atom is shared among 8 adjacent unit cells, so each contributes 1/8 of itself to the unit cell.

DERIVATION: APF FOR SC

1. Number of atoms per unit cell: $Z = 8 \times (1/8) = 1$
2. In SC structure, atoms touch along the edge, so: $2r = a$, which gives $r = a/2$
3. Volume of one atom: $V_{atom} = (4/3)\pi r^3 = (4/3)\pi(a/2)^3 = \pi a^3/6$
4. Volume of unit cell: $V_{cell} = a^3$
5. $APF = (1 \times \pi a^3/6) / a^3 = \pi/6 = 0.5236 \approx 52.36\%$

6.4 APF for Body-Centred Cubic (BCC) Structure

In BCC, atoms are at 8 corners (each shared by 8 cells) plus 1 atom entirely at the body centre.

DERIVATION: APF FOR BCC

1. Number of atoms per unit cell: $Z = 8 \times (1/8) + 1 = 2$
2. In BCC, atoms touch along the body diagonal. Length of body diagonal = $\sqrt{3} \times a$. This diagonal contains 4 radii (corner-body-corner), so: $4r = \sqrt{3}\, a$, giving $r = \sqrt{3}\, a / 4$
3. Volume of 2 atoms: $2 \times (4/3)\pi r^3 = 2 \times (4/3)\pi (\sqrt{3}\, a/4)^3$
4. On simplification: Volume of 2 atoms = $\sqrt{3}\, \pi a^3 / 8$
5. APF = $(\sqrt{3}\, \pi/8) = 0.6802 \approx 68.02\%$

6.5 APF for Face-Centred Cubic (FCC) Structure

In FCC, atoms are at 8 corners (each 1/8) and 6 face centres (each 1/2 shared between 2 cells).

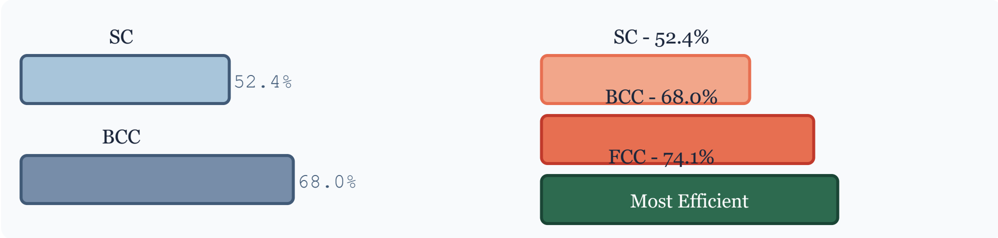
DERIVATION: APF FOR FCC

1. Number of atoms per unit cell: $Z = 8 \times (1/8) + 6 \times (1/2) = 1 + 3 = 4$
2. In FCC, atoms touch along the face diagonal. Face diagonal = $\sqrt{2} \times a$. This contains 4 radii, so: $4r = \sqrt{2}\, a$, giving $r = \sqrt{2}\, a / 4 = a/(2\sqrt{2})$
3. Volume of 4 atoms: $4 \times (4/3)\pi r^3 = 4 \times (4/3)\pi (a/2\sqrt{2})^3$
4. On simplification: Volume of 4 atoms = $\sqrt{2}\, \pi a^3 / 6$
5. APF = $\sqrt{2}\, \pi/6 = 0.7405 \approx 74.05\%$

6.6 Comparison Table: SC vs BCC vs FCC

Property	Simple Cubic (SC)	Body-Centred Cubic (BCC)	Face-Centred Cubic (FCC)
Lattice points / cell	1	2	4
Coordination Number	6	8	12
Atomic Radius (r)	$a/2$	$\sqrt{3}\, a/4$	$\sqrt{2}\, a/4$
Packing Factor (APF)	0.52 (52%)	0.68 (68%)	0.74 (74%)
Void Space	48%	32%	26%
Nearest-neighbour distance	a	$\sqrt{3}\, a / 2$	$a / \sqrt{2}$
Packing type	Loosest	Moderately packed	Most closely packed (cubic)
Examples	Polonium (Po)	Fe, W, Cr, Na, K	Cu, Al, Ag, Au, Ni

Figure 6.1 – Packing efficiency comparison: SC, BCC, FCC



Bar chart showing relative packing efficiency of the three cubic structures

Section 7 | Miller Indices

7.1 Definition

Miller indices are a notation system used to describe the orientation of crystal planes and crystal directions in a lattice. They were introduced by William Hallowes Miller in 1839. Miller indices are a set of three integers (h, k, l) that unambiguously define a family of parallel crystal planes.

7.2 Procedure to Determine Miller Indices

STEP-BY-STEP METHOD FOR FINDING MILLER INDICES OF A PLANE

1. Identify the crystallographic axes (X, Y, Z) of the unit cell.
2. Find the intercepts of the plane on the three axes, in terms of the lattice parameters a, b, c. If a plane is parallel to an axis, its intercept on that axis is infinity (∞).
3. Express the intercepts as multiples of the unit cell dimensions (i.e., divide each intercept by the respective lattice parameter to get pure numbers).
4. Take the reciprocals of these numbers.
5. Convert to the smallest set of whole numbers by multiplying through by the LCM of the denominators (if fractions appear).
6. Write the result as (h k l). A negative index is written with a bar over the number, e.g., ($1\ \bar{1}\ 0$) means h=1, k=-1, l=0.

7.3 Worked Examples

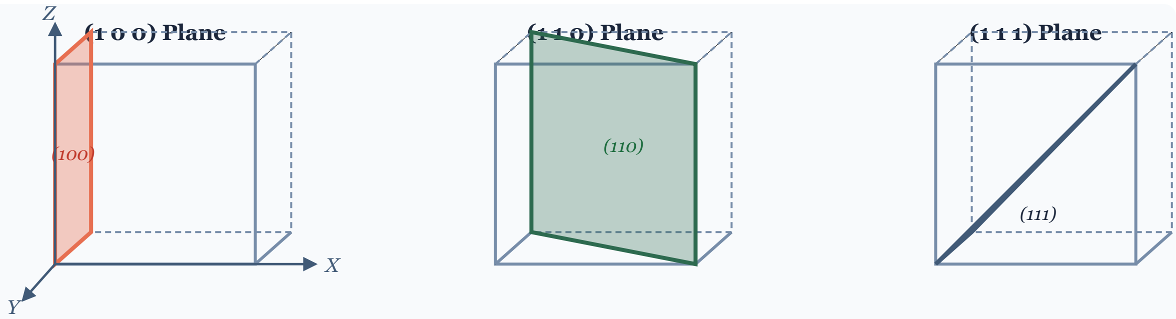
Intercepts on axes	Reciprocals	Miller Indices	Plane Type
1, 1, 1	1, 1, 1	(1 1 1)	Octahedral plane
1, ∞, ∞	1, 0, 0	(1 0 0)	Face plane (perpendicular to X)
1, 1, ∞	1, 1, 0	(1 1 0)	Diagonal plane
1, 2, 3	1, 1/2, 1/3	(6 3 2)	General plane
-1, 1, 1	-1, 1, 1	($\bar{1}\ 1\ 1$)	Negative intercept plane

7.4 Important Properties of Miller Indices

- All parallel planes of the same family have the same set of Miller indices.
- A plane parallel to an axis has a Miller index of 0 for that axis.

- If all indices are multiplied by a common factor n, the resulting indices (nh, nk, nl) represent a plane parallel to (h k l) but at 1/n of the spacing.
- Negative indices indicate that the plane cuts the negative part of the axis.
- The notation {h k l} (curly braces) represents an entire family of equivalent planes by symmetry.
- Square brackets [h k l] are used to denote crystal directions.

Figure 7.1 – Miller index planes (100), (110), and (111) in a cubic crystal



Shaded areas represent the respective crystal planes in a cubic unit cell

Miller index planes (1 0 0), (1 1 0), and (1 1 1) shown as shaded regions within the unit cell

KEY RULES FOR MILLER INDICES

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Miller indices are always integers (after clearing fractions)

A zero index means the plane is parallel to that axis

Family of planes: {h k l}    |    Family of directions: <h k l>

Negative intercept: bar notation ¯h = -h
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Section 8 | Interplanar Spacing

8.1 Definition

The **interplanar spacing** (denoted d_{hkl}) is defined as the perpendicular distance between two adjacent, parallel crystal planes having the same Miller indices (h k l). This spacing is a fundamental parameter in X-ray diffraction, since it directly appears in Bragg's Law.

8.2 Derivation for Cubic Crystal System

For a cubic crystal (a = b = c), the perpendicular distance from the origin to the first (h k l) plane is given by a general geometric argument. Consider a plane with Miller indices (h k l) making intercepts a/h, a/k, and a/l on the X, Y, Z axes respectively.

The unit normal vector to the plane (h k l) is: $\hat{n} = (h\,\hat{\imath} + k\,\hat{\jmath} + l\,\hat{k}) / \sqrt{(h^2 + k^2 + l^2)}$

The perpendicular distance from the origin to this first plane is obtained by taking the dot product of the position vector of any intercept point with the unit normal:

DERIVATION OF INTERPLANAR SPACING D FOR CUBIC SYSTEM

1. A plane with Miller indices (h k l) intercepts the X-axis at a/h. The position vector to this point is (a/h, 0, 0).
2. The unit normal to the (h k l) plane: $\hat{n} = (h, k, l) / \sqrt{(h^2 + k^2 + l^2)}$
3. Interplanar spacing: d = position vector . unit normal = $(a/h) \times (h / \sqrt{(h^2+k^2+l^2)})$
4. Therefore: $d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)}$

INTERPLANAR SPACING FORMULAS

Cubic system: $d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)}$

Tetragonal: $1/d^2 = (h^2 + k^2)/a^2 + l^2/c^2$

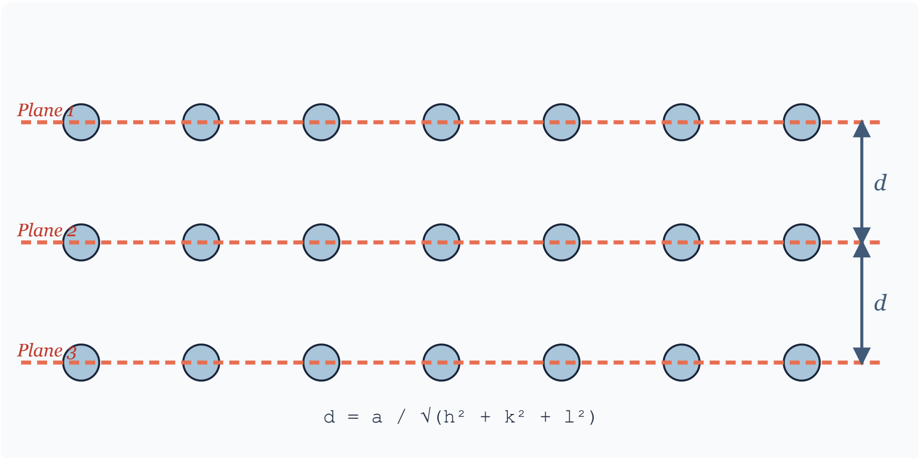
Orthorhombic: $1/d^2 = h^2/a^2 + k^2/b^2 + l^2/c^2$

Hexagonal: $1/d^2 = (4/3)(h^2+hk+k^2)/a^2 + l^2/c^2$

8.3 Values for Common Planes in Cubic Crystal

Plane (h k l)	$h^2+k^2+l^2$	$\sqrt{h^2+k^2+l^2}$	d_{hkl}
(1 0 0)	1	1	a
(1 1 0)	2	√2	a/√2
(1 1 1)	3	√3	a/√3
(2 0 0)	4	2	a/2
(2 1 0)	5	√5	a/√5
(2 1 1)	6	√6	a/√6

Figure 8.1 – Interplanar spacing d_{hkl} between successive (hkl) planes



Successive parallel planes with Miller indices (h k l); d is the perpendicular spacing between adjacent planes

Section 9 | X-ray Diffraction – Overview

9.1 Nature of X-rays

X-rays are electromagnetic radiation with wavelengths in the range of 0.01 Å to 100 Å (1 Å = 10^{-10} m). For crystal structure determination, X-rays with wavelengths comparable to atomic spacings in crystals (approximately 1–2 Å) are used. They are produced by bombarding a metal target (such as Cu, Mo, Fe) with high-energy electrons.

9.2 Why X-rays for Crystal Structure Analysis?

- X-ray wavelengths (0.5–2.5 Å) are of the same order of magnitude as interatomic spacings in crystals (typically 1–5 Å).
- This means crystal planes can act as diffraction gratings for X-rays, producing interference patterns.
- The diffraction pattern carries information about the crystal structure, atomic positions, and unit cell dimensions.
- Visible light ($\lambda \sim 5000$ Å) cannot resolve atomic spacings.

9.3 Phenomenon of Diffraction

When X-rays are incident on a crystal, each atom scatters the radiation in all directions. For most directions, these scattered waves interfere destructively. However, at specific angles satisfying Bragg's Law, constructive interference occurs, resulting in strong diffracted beams. This phenomenon is called **X-ray diffraction (XRD)**.

CONDITION FOR X-RAY DIFFRACTION

Diffraction occurs when the path difference between waves scattered from adjacent crystal planes is an integer multiple of the X-ray wavelength. This is quantified by **Bragg's Law**: $n\lambda = 2d \sin\theta$

9.4 Types of X-ray Diffraction Methods

Method	X-ray Source	Sample Type	Output
Laue Method	Polychromatic (white X-rays)	Single crystal, stationary	Laue spots pattern on flat film
Rotating Crystal Method	Monochromatic	Single crystal, rotated	Spots arranged in layer lines
Powder (Debye-Scherrer) Method	Monochromatic	Polycrystalline powder	Concentric diffraction rings
X-ray Diffractometer	Monochromatic	Flat sample / powder	Intensity vs. 2θ spectrum

Section 10 | Bragg's Law

10.1 Statement of Bragg's Law

Bragg's Law, formulated by William Lawrence Bragg in 1913, states that when a beam of X-rays is incident on a set of crystal planes having interplanar spacing d, constructive interference (maximum diffraction intensity) occurs when the path difference between rays reflected from successive planes is an integer multiple of the wavelength.

BRAGG'S LAW

```
nλ = 2d sinθ

where:

n  = order of diffraction (positive integer: 1, 2, 3, ...)

λ = wavelength of X-rays (in metres or Angstroms)

d  = interplanar spacing of the crystal planes (hkl)

θ = glancing angle (angle between incident X-ray beam and crystal plane)
```

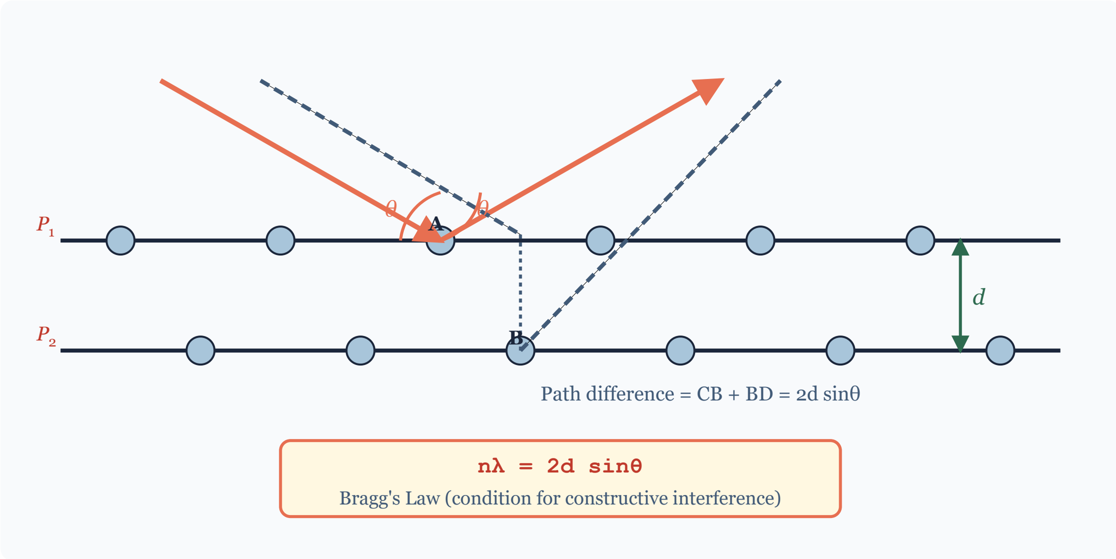
10.2 Derivation of Bragg's Law

Consider two parallel crystal planes P_1 and P_2 with interplanar spacing d. Let an X-ray beam of wavelength λ be incident at a glancing angle θ to the planes. Part of the beam is reflected by plane P_1 at point A, and another part penetrates to plane P_2 and reflects at point B.

STEP-BY-STEP DERIVATION

1. Draw perpendiculars from point A to the incident and reflected rays at B. Let these perpendicular feet be at points C and D on the incident and reflected rays respectively.
2. The path difference between the two rays is: $\Delta = CB + BD$
3. In the triangle ABD: $BD = AB \sin\theta$. Similarly: $CB = AB \sin\theta$.
4. Therefore: Path difference = $CB + BD = AB \sin\theta + AB \sin\theta = 2\ AB \sin\theta$
5. The perpendicular distance AB between the planes equals d, so: Path difference = $2d \sin\theta$
6. For constructive interference, path difference must equal an integer multiple of wavelength: Path difference = $n\lambda$
7. Therefore: $n\lambda = 2d \sin\theta$ (Bragg's Law)

Figure 10.1 – Bragg's Law: X-ray reflection from crystal planes



X-rays incident at glancing angle θ on crystal planes. Constructive interference occurs when path difference = nλ

10.3 Important Points about Bragg's Law

- The angle θ is called the **Bragg angle** or **glancing angle** (not the angle of incidence measured from the normal).
- For diffraction to occur, we need $\lambda \le 2d$ (since $\sin\theta \le 1$).
- For first-order diffraction ($n = 1$): $\lambda = 2d \sin\theta$.
- Each family of planes (h k l) with spacing d_{hkl} diffracts at a unique angle θ_{hkl}.
- From Bragg's Law: if λ is known and θ is measured, d can be calculated, enabling crystal structure determination.

Significance of Bragg's Law

- Foundation of X-ray crystallography
- Allows determination of crystal structure
- Used to measure precise lattice parameters
- Basis for X-ray diffractometry

Limitations of Bragg's Law

- Treats atoms as point scatterers (over-simplified)
- Does not account for atomic form factors
- Assumes perfectly parallel monochromatic X-rays
- Does not explain diffraction intensities

Section 11 | X-ray Diffractometer

11.1 Definition and Purpose

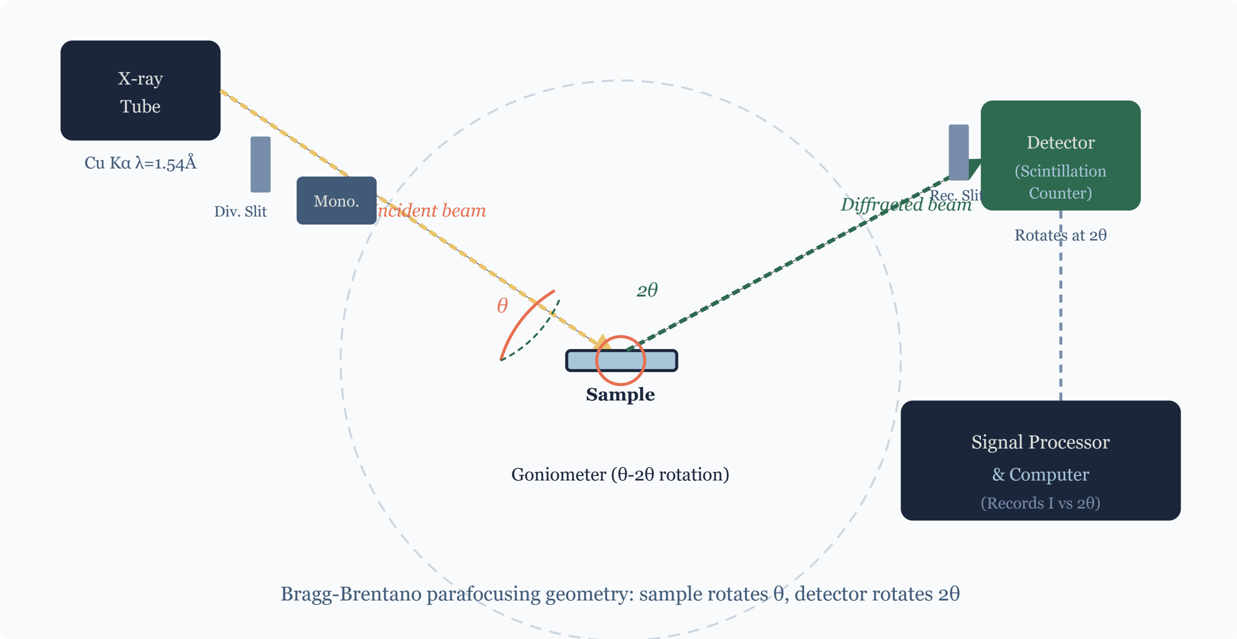
An **X-ray diffractometer** is an instrument that measures the intensity of X-rays diffracted by a crystalline sample as a function of the diffraction angle 2θ. It provides a quantitative diffraction pattern (intensity vs. angle plot) that can be used to identify phases, determine crystal structures, measure lattice parameters, and analyse crystallite sizes and strains.

11.2 Construction and Components

A modern powder X-ray diffractometer consists of the following main components:

- **X-ray Source (X-ray tube):** Generates monochromatic X-rays (usually Cu Kα radiation, λ = 1.5406 Å) by bombarding a copper anode with electrons from a tungsten filament under high voltage (~40 kV).
- **Monochromator / Filter:** A crystal monochromator (e.g., graphite) or Ni filter selects a single wavelength (Kα) from the emitted X-ray spectrum.
- **Divergence Slits and Soller Slits:** Control the angular spread and collimate the X-ray beam before it hits the sample.
- **Sample Stage (Goniometer):** Holds the flat sample (powder or thin film). Rotates precisely by angle θ while the detector simultaneously rotates by 2θ to maintain the Bragg geometry. This mechanism is called the **theta-2theta goniometer**.
- **Anti-scatter Slit and Receiving Slit:** Restrict stray radiation and define the angular resolution of diffracted beams entering the detector.
- **Detector:** Historically a scintillation counter; modern instruments use position-sensitive detectors (PSD) or area detectors that can record multiple angles simultaneously. Converts X-ray photons into electrical pulses.
- **Signal Processing Unit / Computer:** Records intensity as a function of 2θ and produces the diffraction pattern. Software performs phase identification by comparing peak positions with database entries (JCPDS/ICDD database).

Figure 11.1 – Schematic of an X-ray diffractometer (Bragg-Brentano geometry)



Bragg-Brentano geometry: the X-ray tube is fixed; sample rotates by θ while detector rotates by 2θ

11.3 Working Principle

The sample (flat plate of powder) is placed on the goniometer. As the sample rotates by angle θ, the detector simultaneously rotates by 2θ. At angles satisfying Bragg's Law for each set of crystal planes, a peak in intensity is recorded. The resulting **diffractogram** (plot of intensity I vs. 2θ) shows sharp peaks at specific 2θ values corresponding to particular (h k l) planes.

11.4 Applications

- Phase identification of crystalline materials (fingerprinting using JCPDS/ICDD database).
- Determination of lattice parameters with high precision.
- Measurement of crystallite size using the Scherrer equation: $D = K\lambda / (\beta \cos\theta)$, where β is the peak broadening at half-maximum.
- Residual stress and strain analysis in engineering components.
- Quantitative phase analysis in mixtures.
- Thin film analysis (texture, epitaxy, thickness).

Section 12 | Structure Determination: Laue and Powder Methods

12.1 The Laue Method

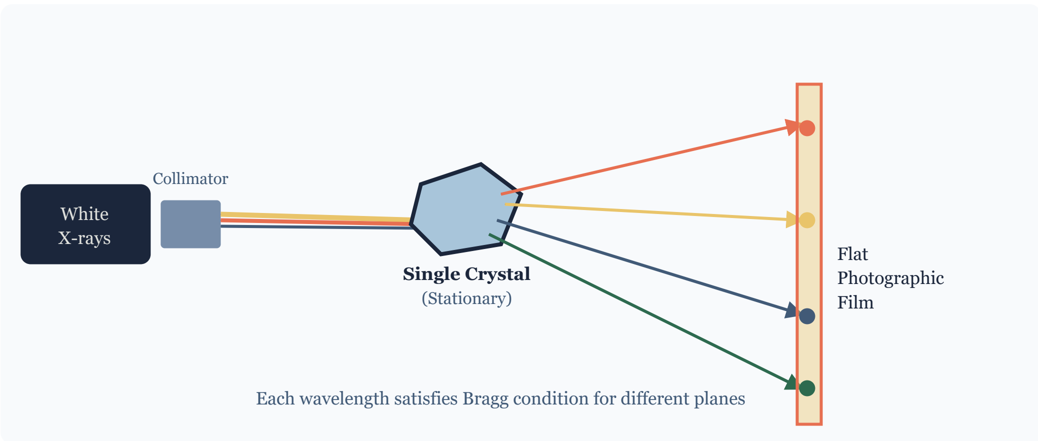
PRINCIPLE

The Laue method uses a **polychromatic (white) X-ray beam** directed at a stationary single crystal. Since the beam contains a continuous range of wavelengths (approximately 0.2 to 2 Å), various planes in the crystal will simultaneously select the correct wavelength to satisfy Bragg's condition. Each set of planes diffracts a different wavelength, and the diffracted beams produce spots on a flat photographic film placed either in front of (transmission Laue) or behind (back-reflection Laue) the crystal.

EXPERIMENTAL SETUP

- A continuous-spectrum (white) X-ray beam is collimated and directed onto a stationary single crystal.
- A flat photographic film is placed perpendicular to the incident beam.
- In transmission Laue, the film is placed beyond the crystal; in back-reflection Laue, in front of it.
- The diffracted spots form a pattern symmetric about the primary beam axis.

Figure 12.1 – Laue method: polychromatic X-rays on stationary single crystal



In the Laue method, each spot on the film corresponds to diffraction from a specific family of planes at a particular wavelength

APPLICATIONS OF LAUE METHOD

- Determination of crystal symmetry and orientation of single crystals.
- Checking the perfection of single crystals.
- Alignment of crystals for neutron/X-ray scattering experiments.

Advantages

- Quick determination of crystal symmetry
- No need to rotate the crystal

Limitations

- Cannot determine lattice parameters precisely
- Spots may overlap for complex structures

- Useful for oriented single crystals

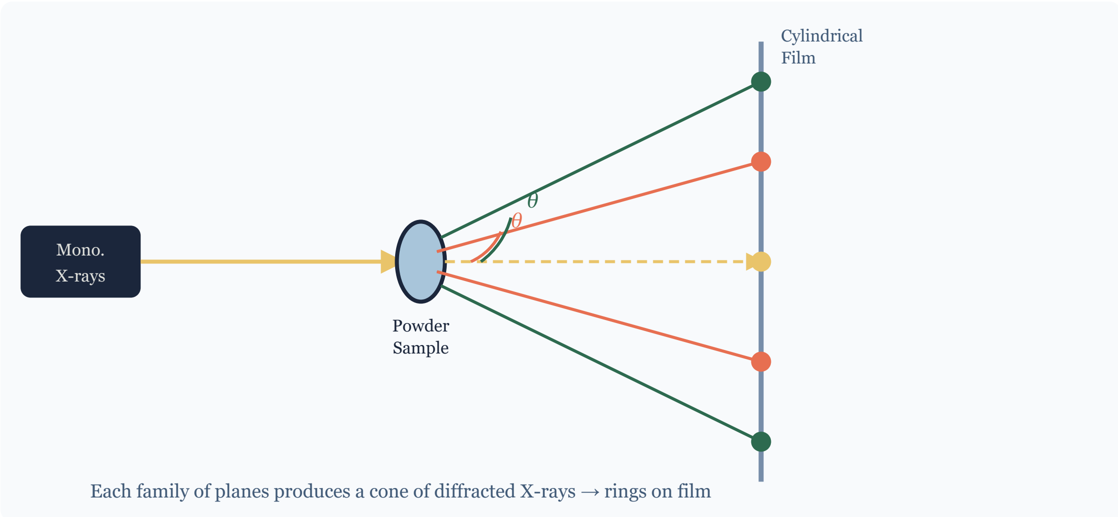
- Needs a large single crystal

12.2 The Powder Method (Debye-Scherrer Method)

PRINCIPLE

The powder method uses **monochromatic X-rays** incident on a finely powdered (polycrystalline) sample. Because the powder contains a very large number of tiny crystallites with random orientations, for every family of crystal planes (h k l), some crystallites will be oriented at exactly the Bragg angle θ relative to the incident beam. These crystallites diffract the X-rays, producing a **cone of diffracted radiation** for each (h k l) plane family. When recorded on cylindrical film or a flat detector, these cones appear as **concentric arcs or rings** (Debye-Scherrer rings).

Figure 12.2 – Powder (Debye-Scherrer) method: monochromatic X-rays on powdered sample



Debye-Scherrer powder method: each set of planes at Bragg angle produces a diffraction cone recorded as rings

PROCEDURE AND ANALYSIS

STEPS IN POWDER METHOD CRYSTAL STRUCTURE DETERMINATION

1. Grind the material to a fine powder and mount it as a thin cylinder in the sample holder.
2. Expose the sample to monochromatic X-rays of known wavelength λ.
3. Record the diffracted rings on cylindrical photographic film or area detector.
4. Measure the positions (2θ values) of the diffraction rings/peaks.
5. Use Bragg's Law to calculate d-spacings: $d = \lambda / (2 \sin\theta)$ for each peak.
6. Index the peaks by assigning Miller indices (h k l) to each reflection based on the d-spacing ratios.
7. Calculate the lattice parameter from: $a = d \sqrt{(h^2 + k^2 + l^2)}$ (for cubic).

12.3 Comparison: Laue vs Powder Method

Parameter	Laue Method	Powder (Debye-Scherrer) Method
X-ray type	Polychromatic (white X-rays)	Monochromatic
Sample type	Single crystal (stationary)	Polycrystalline powder
Variable	Wavelength λ varies; θ is fixed	Angle θ varies; λ is fixed
Output pattern	Spots on flat film	Concentric rings on cylindrical film
Main use	Crystal orientation and symmetry	Lattice parameter determination and phase ID
Precision	Low (no precise d values)	High (precise lattice parameters)
Sample requirement	Large single crystal needed	Micro-gram quantities sufficient
Speed	Fast (simultaneous diffraction)	Slower, but modern diffractometers are fast

Section 13 | Questions and Answers

Part A – 2-Mark Questions

Q1. Define a space lattice. 2 marks

A **space lattice** is an infinite, three-dimensional periodic arrangement of geometric points such that every point has an identical environment. Every lattice point can be reached from the origin by the lattice vector $R = n_1a + n_2b + n_3c$, where n_1, n_2, n_3 are integers and a, b, c are primitive translation vectors. There are exactly 14 distinct space lattices (Bravais lattices) in three dimensions. The space lattice represents the geometric framework of a crystal without specifying the actual atoms present.

Q2. What is a unit cell? Define lattice parameters. 2 marks

A **unit cell** is the smallest repeating structural unit of a crystal that, when translated in three dimensions, generates the complete crystal. It is characterised by six lattice parameters: three edge lengths a, b, c and three inter-axial angles α (between b and c axes), β (between a and c axes), and γ (between a and b axes). These six parameters completely define the size and shape of the unit cell and determine which of the seven crystal systems the material belongs to.

Q3. State Bragg's Law and define the terms used. 2 marks

Bragg's Law states that when X-rays are incident on a crystal at a glancing angle θ, constructive diffraction occurs when: $\mathbf{n\lambda = 2d \sin\theta}$, where n is the order of diffraction (positive integer), λ is the X-ray wavelength, d is the interplanar spacing between crystal planes (h k l), and θ is the glancing angle between the incident beam and the crystal plane. This condition ensures that the path difference between rays reflected from successive parallel planes equals an integer number of wavelengths, producing constructive interference.

Q4. Define Miller indices. How is a plane with negative intercept denoted? 2 marks

Miller indices are a set of three smallest possible integers (h, k, l) used to specify the orientation of a crystal plane. They are obtained by finding the intercepts of the plane on the three crystallographic axes, taking reciprocals, and clearing fractions. A plane with a negative intercept on an axis is denoted by placing a bar over the corresponding index, written as $\bar{h}$, $\bar{k}$, or $\bar{l}$. For example, a plane cutting the positive X-axis, negative Y-axis, and positive Z-axis would be written as $(1\,\bar{1}\,1)$. Parallel planes of the same family share the same Miller indices.

Q5. What is the coordination number of atoms in BCC and FCC structures? 2 marks

The **coordination number** is the number of nearest atoms directly touching a given atom. In a **Body-Centred Cubic (BCC)** structure, the body-centre atom has 8 nearest neighbours at the corners of the cube, so the coordination number is 8. In a **Face-Centred Cubic (FCC)** structure, each atom is surrounded by 12 nearest neighbours — 4 in the same layer, 4 in the layer above, and 4 in the layer below — giving a coordination number of 12. A higher coordination number implies greater packing efficiency (FCC has APF = 0.74, BCC has APF = 0.68).

Q6. Write the formula for interplanar spacing in a cubic crystal. 2 marks

For a cubic crystal system with lattice parameter a, the interplanar spacing d between adjacent parallel planes with Miller indices (h k l) is given by: $\mathbf{d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)}}$. From this formula, we can see that planes with smaller Miller indices (smaller $h^2+k^2+l^2$) have larger interplanar spacings. For example: $d_{100} = a$, $d_{110} = a/\sqrt{2}$, and $d_{111} = a/\sqrt{3}$. The ratio $d_{100} : d_{110} : d_{111} = 1 : (1/\sqrt{2}) : (1/\sqrt{3})$, which is a characteristic signature of a cubic crystal.

Part B – 5-Mark Questions

Q7. Explain Bragg's Law with derivation and a neat diagram. 5 marks

Statement: When a monochromatic X-ray beam of wavelength λ is incident at a glancing angle θ on a set of crystal planes with interplanar spacing d, constructive diffraction occurs when $n\lambda = 2d \sin\theta$ (Bragg's Law), where n is a positive integer called the order of diffraction.

Derivation: Consider parallel crystal planes P_1 and P_2 separated by distance d. An X-ray beam is incident at glancing angle θ. Part of the beam reflects from plane P_1 at point A, while the remaining beam penetrates to plane P_2 and reflects at point B. Drop perpendiculars from A to the incident and reflected rays at B, meeting them at points C and D. The extra path travelled by the second beam is CB + BD. In right triangle CAB: $CB = AB \sin\theta$. Similarly, $BD = AB \sin\theta$. Since AB = d (separation between planes), the total path difference = $2d \sin\theta$.

For constructive interference, this path difference must equal n wavelengths: $\mathbf{n\lambda = 2d \sin\theta}$.

Significance: (i) If λ is known and θ is measured experimentally, d can be calculated to determine crystal structure. (ii) Since $\sin\theta \le 1$, the condition $\lambda \le 2d$ must be satisfied for diffraction to occur. (iii) Each (h k l) family diffracts at a unique angle, producing a characteristic diffraction pattern. (iv) First-order diffraction (n = 1) gives the strongest peak. (v) Bragg's Law is the cornerstone of X-ray crystallography, electron diffraction, and neutron diffraction techniques.

Q8. Define Miller indices. Explain the procedure for determining Miller indices of a plane. Give examples. 5 marks

Definition: Miller indices are the smallest set of three integers (h, k, l) that describe the orientation of a crystal plane. Named after W.H. Miller (1839), they are derived from the intercepts of the plane on the three crystallographic axes.

Procedure: Step 1: Note the intercepts of the plane on the X, Y, Z axes in terms of the lattice parameters a, b, c. If the plane is parallel to an axis, its intercept is infinity. Step 2: Divide by the lattice parameters to get dimensionless intercepts. Step 3: Take the reciprocals. Step 4: Multiply by the LCM of all denominators to obtain the smallest integers. Step 5: Write the result enclosed in parentheses (h k l) without commas.

Examples: (i) A plane cutting X at 1, Y at infinity, Z at infinity: reciprocals = 1, 0, 0 → Miller indices (1 0 0). (ii) A plane cutting X at 1, Y at 1, Z at infinity: reciprocals = 1, 1, 0 → Miller indices (1 1 0). (iii) A plane cutting X at 1, Y at 2, Z at 3: reciprocals = 1, 1/2, 1/3; LCM = 6; multiply: 6, 3, 2 → Miller indices (6 3 2).

Important properties: (i) Parallel planes have the same Miller indices. (ii) Curly braces {h k l} denote a family of equivalent planes. (iii) Square brackets [h k l] denote a crystal direction. (iv) A negative intercept gives a bar notation. (v) Miller indices are used in the interplanar spacing formula and in X-ray diffraction analysis to index diffraction peaks.

Q9. Compare SC, BCC, and FCC crystal structures in terms of coordination number, packing factor, and number of atoms per unit cell. Derive APF for any one structure. 5 marks

The three cubic crystal structures differ fundamentally in the arrangement and number of atoms per unit cell.

Simple Cubic (SC): 8 corner atoms each contributing 1/8 to the cell → 1 atom/cell. Nearest neighbours are along face edges → CN = 6. Atoms touch along the cube edge: $2r = a \rightarrow r = a/2$. APF = $(4\pi r^3/3)/a^3 = \pi/6 = 0.52$ (52%).

BCC: 8 corners (1 atom) + 1 body centre = 2 atoms/cell. Body-centre atom touches corner atoms along body diagonal → CN = 8. $4r = \sqrt{3}a \rightarrow r = \sqrt{3}a/4$. APF = $2(4\pi r^3/3)/a^3 = \sqrt{3}\pi/8 = 0.68$ (68%).

FCC: 8 corners (1) + 6 face centres (3) = 4 atoms/cell. Each face-centre atom has 4 face-centre and 8 corner neighbours → CN = 12. $4r = \sqrt{2}a \rightarrow r = a/(2\sqrt{2})$. APF = $4(4\pi r^3/3)/a^3 = \sqrt{2}\pi/6 = 0.74$ (74%).

Comparison: FCC has the highest packing efficiency (74%) and coordination number (12), making it the most closely packed cubic structure. BCC materials (Fe, W, Cr) have good mechanical properties. FCC materials (Cu, Al, Au) are highly ductile. SC (only Po) is the least stable due to low packing. The sequence in packing efficiency is: SC (52%) < BCC (68%) < FCC (74%).

Q10. Describe the Powder (Debye-Scherrer) method of crystal structure determination. How is data analysed to find lattice parameters? 5 marks

Principle: The powder method uses monochromatic X-rays incident on a finely powdered polycrystalline sample. Since the powder contains randomly oriented crystallites, for every (h k l) plane family, some crystals will be oriented at the correct Bragg angle θ. These diffract the X-rays into a cone of radiation, producing rings on a surrounding cylindrical film.

Experimental setup: (i) The sample is ground to a fine powder and loaded into a thin-walled capillary tube or stuck to a fiber as a thin cylinder. (ii) The sample is placed at the centre of a cylindrical camera surrounded by a strip of photographic film. (iii) A collimated beam of monochromatic X-rays is directed at the sample. (iv) After sufficient exposure, the film is developed, showing a series of concentric arcs (Debye-Scherrer rings) symmetric about the primary beam direction.

Analysis: Step 1: Measure the positions (S, in mm) of symmetric arc pairs on the developed film. Step 2: Calculate 2θ from S using: $2\theta = S/R$ (radians), where R is the camera radius. Step 3: Apply Bragg's Law to find d-spacings: $d = \lambda/(2 \sin\theta)$. Step 4: Calculate $h^2+k^2+l^2 = a^2/d^2$ for cubic system. Step 5: The ratio values identify the crystal structure (SC, BCC, FCC have distinct allowed reflections). Step 6: The lattice parameter $a = d \sqrt{(h^2+k^2+l^2)}$ is calculated and averaged over all peaks for precision.

Applications: Phase identification, precise lattice parameter determination, crystallite size estimation via peak broadening, stress and strain analysis, identification of unknowns. The powder method is the most widely used XRD technique due to its simplicity and applicability to any polycrystalline material.

Q11. Explain the X-ray diffractometer with a labelled diagram. How is it used to determine crystal structure? 5 marks

Definition: An X-ray diffractometer is an instrument that measures the diffracted X-ray intensity as a function of angle 2θ using Bragg-Brentano geometry, producing a diffractogram (I vs 2θ) that carries complete structural information about the sample.

Main components: (i) X-ray tube (typically Cu anode, λ = 1.5406 Å), (ii) monochromator or beta filter to select Kα radiation, (iii) divergence and soller slits to collimate the beam, (iv) goniometer stage with rotating sample and detector, (v) anti-scatter and receiving slits, (vi) scintillation or semiconductor detector, and (vii) computer for data acquisition and analysis.

Working: The flat sample is mounted on the goniometer. As the sample rotates by angle θ and the detector rotates by 2θ simultaneously, the detector records the intensity of diffracted X-rays. When the angle satisfies Bragg's condition for a particular (h k l) plane, a sharp peak appears in the diffractogram. The full scan from 2θ = 5° to 90° generates a complete diffraction pattern.

Crystal structure determination: (i) Identify peak positions (2θ). (ii) Calculate d-spacings using d = λ/(2 sinθ). (iii) Index peaks by comparing d-spacing ratios to theoretical values for cubic (1:1/√2:1/√3...) or other systems. (iv) Compute lattice parameter a. (v) Compare with JCPDS database to identify phase. (vi) Peak intensities give information about atom positions within the cell via structure factor analysis.

Section 14 | Quick Revision Sheet

Unit II – Crystallography and X-ray Diffraction: At a Glance

KEY DEFINITIONS

- Space Lattice: Infinite periodic array of points in 3D
- Basis: Atom/group attached to each lattice point
- Crystal = Lattice + Basis
- Unit Cell: Smallest repeating structural unit
- 14 Bravais Lattices in 7 Crystal Systems

ESSENTIAL FORMULAS

Lattice vector: $R = n_1a + n_2b + n_3c$

$APF = (Z \times 4\pi r^3/3) / a^3$

$d_{pnm} = a / \sqrt{(h^2+k^2+l^2)}$ [cubic]

Bragg's Law: $n\lambda = 2d \sin\theta$

CUBIC STRUCTURES: SC | BCC | FCC

- Atoms/cell: 1 | 2 | 4
- CN: 6 | 8 | 12
- APF: 0.52 | 0.68 | 0.74
- r: $a/2$ | $\sqrt{3}a/4$ | $a/2\sqrt{2}$
- Touch along: edge | body diag | face diag

MILLER INDICES: STEPS

- 1. Find intercepts on X, Y, Z axes
- 2. Express in terms of a, b, c
- 3. Take reciprocals
- 4. Clear fractions to integers
- 5. Write as (h k l)
- Parallel to axis → index = 0

SEVEN CRYSTAL SYSTEMS

- Cubic: $a=b=c$, $\alpha=\beta=\gamma=90°$
- Tetragonal: $a=b\neq c$, all 90°
- Orthorhombic: $a\neq b\neq c$, all 90°
- Hexagonal: $a=b\neq c$, $\gamma=120°$
- Rhombohedral: $a=b=c$, $\alpha=\beta=\gamma\neq 90°$
- Monoclinic: $a\neq b\neq c$, $\beta\neq 90°$
- Triclinic: $a\neq b\neq c$, $\alpha\neq\beta\neq\gamma\neq 90°$

XRD METHODS COMPARISON

- Laue: White X-rays, single crystal, stationary
- Laue: Gives symmetry and orientation
- Powder: Mono. X-rays, polycrystalline
- Powder: Gives lattice parameters
- Diffractometer: I vs 2θ scan, phase ID

BRAGG'S LAW SUMMARY

$n\lambda = 2d \sin\theta$

- n = order (1, 2, 3...)
- λ = X-ray wavelength
- d = interplanar spacing
- θ = glancing angle
- Condition: $\lambda \leq 2d$
- 1st order strongest; higher n weaker

IMPORTANT D-SPACINGS (CUBIC)

- (100): $d = a$
- (110): $d = a/\sqrt{2} \approx 0.707a$
- (111): $d = a/\sqrt{3} \approx 0.577a$
- (200): $d = a/2$
- (210): $d = a/\sqrt{5} \approx 0.447a$
- (211): $d = a/\sqrt{6} \approx 0.408a$

Key Equations Summary

Bragg's Law: $n\lambda = 2d \sin\theta$

Interplanar spacing [cubic]: $d = a / \sqrt{(h^2 + k^2 + l^2)}$

APF (SC): $\pi/6 = 0.5236$

APF (BCC): $\sqrt{3}\pi/8 = 0.6802$

APF (FCC): $\sqrt{2}\pi/6 = 0.7405$

Scherrer equation: $D = K\lambda / (\beta \cos\theta)$

Lattice vector: $R = n_1a + n_2b + n_3c$

Section 15 | Textbooks and References

Textbooks

- **Avadhanulu, M.N., Kshirsagar, P.G., and Arun Murthy, T.V.S.** – *A Textbook of Engineering Physics*. S. Chand and Company, New Delhi, 2019. [Primary reference for Unit II crystallography and X-ray diffraction topics]
- **Bhattacharya, D.K. and Tandon, Poonam** – *Engineering Physics*. Oxford University Press, New Delhi, 2015. [Covers Bragg's Law derivation, diffractometer setup, and powder method in detail]

Reference Books

- **Pandey, B.K. and Chaturvedi, S.** – *Engineering Physics*. Cengage Learning India, 2021. [Advanced treatment of crystal systems and Bravais lattices]
- **Sharma, Shatendra and Sharma, Jyotsna** – *Engineering Physics*. Pearson Education India, 2018. [Good coverage of Miller indices and X-ray diffraction methods]
- **Jain, Sanjay D. et al.** – *Engineering Physics*. Universities Press, Hyderabad, 2010. [Useful for worked numerical problems on packing factors and interplanar spacing]
- **Srinivasan, M.R.** – *Engineering Physics*. New Age International Publishers, New Delhi, 2009. [Clear explanations of Laue and powder methods]

Standard Reference Works

- **Kittel, Charles** – *Introduction to Solid State Physics*, 8th edition. John Wiley and Sons, 2004. [Definitive text for advanced crystal structure and X-ray diffraction]
- **Hammond, C.** – *The Basics of Crystallography and Diffraction*, 4th edition. Oxford University Press, 2015. [Excellent for understanding Laue patterns and powder diffractometry]
- **Cullity, B.D. and Stock, S.R.** – *Elements of X-ray Diffraction*, 3rd edition. Prentice Hall, 2001. [Industry-standard reference for XRD analysis]

Note for Students: For B.Tech first-year examination, focus primarily on Avadhanulu & Kshirsagar and D.K. Bhattacharya. The derivations in Sections 6 (APF) and 10 (Bragg's Law) are frequently asked in university examinations and must be practised thoroughly. The comparison tables and formula boxes in this module are designed to aid quick revision.

Content prepared in alignment with university curriculum. All formulas verified against standard textbook references listed in Section 15.

— End of Module —